



National University
of Computer & Emerging Sciences Islamabad



MT1008 – Multivariable Calculus

Assignment No: 04

Marks: 120

Section: BCS (A to J)

CLO: 03

Deadline: 21-04-2026 before 02:00pm

Semester: Spring-26

Instructions:

1. Plagiarized work will result in zero marks.
 2. No retake or late submission will be accepted.
 3. Attach complete code, results, and screenshot for questions that require programming solutions. Programs/codes should be typed.
 4. **Complete assignment is to be submitted in soft-copy as well as in hard-copy. Submit the hardcopy before the deadline through CR, and softcopy on GCR.**
 5. The softcopy should be a single PDF file of your complete assignment including programming and non-programming questions.
 6. The PDF file should be according to the following format: id_section_A1 e.g., i25123456_A_A2. A2 in the end denotes Assignment 2. **The title page must include complete student information, including name, section, id, course name, and assignment number.**
 7. The images of a by-hand solution should be properly scanned. You can use any mobile application such as CamScanner or Adobe Scan for scanning. Each of these applications allows you to export pdf or image files which you can use to combine with your programming solutions. Do not attach direct images from the camera application of your mobile phone, or screenshots.
 8. Python is the only approved programming language.
-

Problem 1: Cartesian Coordinates (20 Marks)

A computing system distributes workload over a region. The load intensity at any point (x,y) is given by: $f(x,y) = x^2y + \sin(xy)$ The region R is bounded by:

- $y = x$
- $y = 2x$
- $x = 2$

Tasks:

(a) (4 Marks) Sketch the region R , identify intersection points, and describe its shape.

(b) (4 Marks) Set up the double integral

$$\iint_R f(x,y) dA$$

in both orders: $(dy dx)$ and $(dxdy)$.

(c) (3 Marks) Which order of integration is easier? Explain your reasoning based on the function.

(d) (6 Marks) Evaluate the double integral $\iint_R f(x,y) dA$

(e) (3 Marks) Compute the average value of the function over the region, and explain what this represents in the context of system performance.

Problem 2: Polar Coordinates (20 Marks)

A city has a circular layout, and its population density varies with distance from the center. The density at any point (x,y) (people/km²) is:

$$f(x,y) = 4000 e^{-0.01(x^2+y^2)}$$

The city covers the region:

$$x^2 + y^2 \leq 49$$

Tasks:

(a) (4 Marks) Explain why polar coordinates are best suited here. Convert $f(x,y)$ to polar form and rewrite the region using (r,θ) .

(b) (5 Marks) Set up the double integral in polar coordinates to represent the total population:

$$P = \iint_R f(r, \theta) r \, dr \, d\theta$$

(c) (6 Marks) Evaluate the integral to compute the total population.

(d) (5 Marks) Write a Python program to approximate the total population using a polar grid (discretize r and θ).

- Compare the numerical approximation to the exact value.
- Investigate the effect of increasing the number of grid points.

Problem 3: Volume & Surface Area (20 Marks)

A new auditorium has a **quarter-circular foundation** of radius 50 ft:

$$x^2 + y^2 = 50^2, \quad x \geq 0, \quad y \geq 0$$

The floor and ceiling are modeled by:

Floor: $z = \frac{1}{4}(x + 2y)$

Ceiling: $z = 15 + \frac{x^2 + y^2}{100}$

Tasks:

(a) (6 Marks) Set up and evaluate a triple integral to calculate the total volume of the auditorium.

(b) (6 Marks) Compute the surface area of the ceiling using:

$$S = \iint_R \sqrt{1 + \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2} \, dA$$

(c) (8 Marks) Write a Python program to:

- Approximate the volume numerically.
- Approximate the ceiling surface area numerically.

Problem 4: Maximum Energy/Density Region (20 Marks)

In a physical system, the **energy density** at any point (x, y, z) in space is given by

$$\rho(x,y,z) = 1 - x^2 - y^2 - z^2.$$

This model describes a situation where energy is highest at the center and decreases outward.

(a) (5 Marks) Determine the solid region Q in space where the energy density is **physically meaningful (non-negative)**.

(b) (5 Marks) Interpret the triple integral

$$\iiint_Q (1 - x^2 - y^2 - z^2) \, dV$$

in the context of this problem. What physical quantity does it represent?

(c) (5 Marks) Evaluate the total energy contained in the region Q using an appropriate coordinate system.

(d) (5 Marks) Use a computer algebra system (CAS / Python / MATLAB) to approximate the total energy.

Problem 5: Cylindrical Coordinates (20 Marks)

A storage tank is shaped as a **spherical dome** on top and a **Conical base** below:

$$\text{Dome: } x^2 + y^2 + z^2 = 16, \quad \text{Cone: } z = k\sqrt{x^2 + y^2}$$

The tank contains a liquid of **uniform density**.

Tasks:

(a) (4 Marks) Express the surfaces in cylindrical coordinates (r, θ, z) and describe the region of the liquid. Determine limits of integration.

(b) (6 Marks) Set up and evaluate a triple integral in cylindrical coordinates to find the total volume.

(c) (6 Marks) Assuming constant density $\rho = 1$, determine the center of mass:

$$\bar{x} = \frac{1}{M} \iiint x \, dV, \quad \bar{y} = \frac{1}{M} \iiint y \, dV, \quad \bar{z} = \frac{1}{M} \iiint z \, dV, \quad M = \iiint dV$$

(d) (4 Marks) Write a program to:

- Numerically approximate the volume.
- Approximate the z -coordinate of the center of mass.

Problem 6: Spherical Coordinates (20 Marks)

A spherical shielding system in a reactor occupies the region:

$$a^2 \leq x^2 + y^2 + z^2 \leq b^2, \quad b > a$$

Only the portion inside the cone $z^2 = x^2 + y^2$ is used. Material density is proportional to distance from the origin:

$$\rho(x,y,z) = k\sqrt{x^2 + y^2 + z^2}$$

Tasks:

- (a) (4 Marks) Convert to spherical coordinates (ρ, θ, ϕ) and describe the region. Determine limits of integration.
- (b) (6 Marks) Evaluate the volume of the shell using spherical coordinates.
- (c) (6 Marks) Compute the total mass using the given density function.
- (d) (4 Marks) Using a CAS / Python, approximate volume and mass numerically. Use sample values: $a = 1, b = 2, k = 1$.